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# HiLight: A Hierarchical Reinforcement Learning Framework with Global Adversarial Guidance for Large-Scale Traffic Signal Control

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## Abstract

Efficient traffic signal control (TSC) is essential for mitigating urban congestion, yet existing reinforcement learning (RL) methods face challenges in scaling to large networks while maintaining global coordination. Centralized RL suffers from scalability issues, while decentralized approaches often lack unified objectives, resulting in limited network-level efficiency. In this paper, we propose HiLight, a hierarchical reinforcement learning framework with global adversarial guidance for large-scale TSC. HiLight consists of a high-level Meta-Policy, which partitions the traffic network into subregions and generates sub-goals using a Transformer-LSTM architecture, and a low-level Sub-Policy, which controls individual intersections with global awareness. To improve the alignment between global planning and local execution, we introduce an adversarial training mechanism, where the Meta-Policy generates challenging yet informative sub-goals, and the Sub-Policy learns to surpass these targets, leading to more effective coordination. We evaluate HiLight across both synthetic and real-world benchmarks, and additionally construct a large-scale Manhattan network with diverse traffic conditions, including peak transitions, adverse weather, and holiday surges. Experimental results show that HiLight exhibits significant advantages in large-scale scenarios and remains competitive across standard benchmarks of varying sizes.

## 1 Introduction

Urban traffic congestion has become one of the most pressing challenges in modern cities, leading to increased travel times, environmental pollution, and economic losses. As urban populations continue to grow, efficient traffic management systems are essential for maintaining the functionality of urban infrastructure. Among various solutions, TSC plays a central role in optimizing vehicle flows at intersections to mitigate congestion. Traditional TSC methods, such as Fixed-Time Control (FTC)[1], MaxPressure control[2], and adaptive systems like SCOOT[3], have been widely deployed in real-world systems. However, these approaches often rely on static configurations or pre-defined heuristics, making them ill-suited for coping with highly dynamic and stochastic traffic conditions that frequently occur in urban environments.

In recent years, RL has emerged as a promising approach for adaptive TSC[4, 5, 6], owing to its ability to learn optimal policies through interaction with the environment. As illustrated in Figure 1b, existing RL-based TSC methods are broadly categorized into centralized and multi-agent approaches. Centralized RL treats the entire network as a single decision-maker and effectively optimizes global objectives[7, 8, 9], but suffers from poor scalability due to the exponential growth of joint state and action spaces. Multi-agent RL improves scalability by assigning an agent to each intersection[10, 11, 12, 13], yet often lacks global coordination, leading to misaligned local decisions and reduced system-wide efficiency. **Challenge 1: How to develop an RL-based TSC method that simultaneously ensures scalability and global coordination remains unresolved.**

Despite the development of various RL algorithms, the evaluation of TSC methods remains constrained by limitations of existing benchmark scenarios. Widely used scenarios (such as Grid5×5[14], Arterial4×4[15], and Cologne8[16]) are based on either synthetic road networks or simplified real-world maps, and the traffic flows are simulated using default configurations or real data sampled from a narrow time window. **Challenge 2: These scenarios fail to capture the complexity and variability of real-world traffic,** which can be influenced by diverse conditions such as adverse weather, holiday travel surges, or transitions between peak and off-peak periods, as shown in Figure 1a.

Several non-trivial technical issues must be resolved when addressing **these two challenges**. First, agents must optimize global rather than local traffic objectives, which requires capturing long-range dependencies and coordinating actions across distant intersections. Second, preserving global information in large networks raises scalability concerns, as the observation space expands rapidly with the number of intersections, leading to memory and computational bottlenecks. Third, coordinating a growing number of agents becomes increasingly complex. Additionally, it is essential to construct diverse traffic flow settings that reflect real-world conditions, enabling a more comprehensive evaluation of TSC methods.

To tackle these issues, we propose the HiLight, which is a hierarchical RL framework consists of two components: Meta-Policy and Sub-Policy. The high-level Meta-Policy partitions the traffic network into subregions and generates sub-goal using an architecture that combines a Transformer encoder and LSTM, to effectively capture spatial correlations and temporal dynamics. The sub-goal is then used to guide the low-level Sub-Policy agents, which control individual intersections and make fine-grained decisions based on their local observations, as well as global features provided by the Meta-Policy. In addition, an adversarial training strategy is employed, in which the Meta-Policy learns to produce increasingly challenging sub-goals, while the Sub-Policy is trained to outperform these goals in actual traffic performance. To evaluate the effectiveness of the proposed method across different conditions, we conduct experiments in both large-scale and standard benchmark scenarios. Specifically, we design three diverse traffic flow patterns based on the Manhattan network to serve as large-scale stress tests, and also validate our method on five widely used traffic environments. Experimental results demonstrate that our approach achieves strong and consistent performance, particularly under large-scale and dynamic urban traffic settings.

## 2 Related Works

**Traffic Signal Control** TSC involves dynamically adjusting traffic light phases to optimize flow across a road network. Traditional methods—such as FTC[1], MaxPressure[2], SCOOT[3], and SCATS[17]—rely on heuristic rules or local feedback, limiting their adaptability to dynamic and uncertain traffic conditions. Subsequent studies explored kernel-based methods and natural actor-critic algorithms to enhance function approximation capabilities, and adopted deep reinforcement learning techniques such as DQN, DDPG, and A2C to improve policy learning [18, 19, 20, 21, 22]. However, these methods are typically evaluated in isolated or small-scale networks and fail to address scalability in large-scale scenarios.

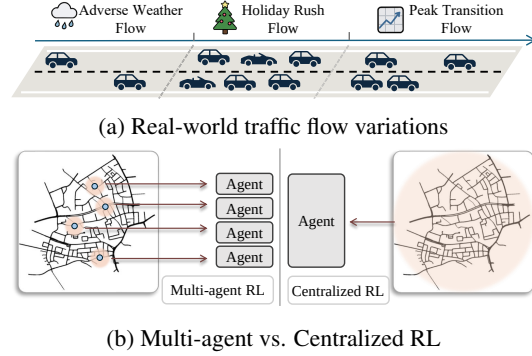


Figure 1: Realistic traffic flow scenarios and RL architectures for TSC

Multi-agent reinforcement learning (MARL) approaches, such as IDQN [23, 24], IPPO [25], and MAPPO [26], often rely on agent interaction only during training, leading to poor collaboration during decentralized execution. To enhance coordination among intersections, recent studies introduced explicit collaboration mechanisms, such as modeling neighboring representations using graph neural networks (e.g., LibSignal, CoLight [27], MaCAR [12], DynSTGAT [28]) or applying clustering strategies to form cooperative groups [29, 13]. However, these methods typically enable collaboration only within limited regions and fail to provide global context to all intersections in large-scale networks, thereby overlooking the potential inconsistency between local optimization and global traffic efficiency. In addition, MetaLight [30] and MetaVIM [31] adopt meta-learning techniques to improve generalization across scenarios, yet their reliance on complex architectures or handcrafted rules hampers scalability and deployment in heterogeneous real-world traffic networks.

**Hierarchical RL** Hierarchical reinforcement learning (HRL) enables more efficient learning and decision-making by structuring policies across multiple levels, where higher-level controllers make decisions and lower-level policies execute fine-grained actions. Existing HRL methods can be broadly categorized into options-based frameworks[32, 33], goal-conditioned hierarchical policies[34, 35], and feudal or manager-worker architectures[36, 37]. These approaches commonly define a high-level policy (or manager) that selects sub-goals or tasks, while a low-level policy (or worker) executes primitive actions to achieve them. Although HRL has demonstrated strong performance in domains such as robotics and navigation, it has rarely been applied to large-scale urban traffic control, where the agent must coordinate across many regions, balance local reactivity with global efficiency, and adapt to diverse traffic patterns.

### 3 Problem Definition

#### 3.1 Preliminaries

**Traffic Intersection** Intersection refers to a location in the road network where two or more roads cross or merge, typically equipped with traffic signals to regulate vehicle movement. Figure 2 shows a typical example: a standard four-leg intersection (also called a four-arm intersection), where four roads meet, forming a cross shape. Each approach (arm) contains three incoming lanes  $l_{in}$  dedicated to left-turn, through, and right-turn movements, and typically three corresponding outgoing lanes  $l_{out}$ .

**Signal Phase** A phase is defined as the set of green light indications for all incoming lanes at an intersection. As illustrated in Figure 2, a standard intersection typically adopts eight phases, each governing vehicle movements such as left turns, through movements, and right turns. In some scenarios, a single incoming lane may accommodate multiple movement directions. For example, in the Arterial 4x4 scenario, certain lanes serve both through and right-turning vehicles. Additionally, in accordance with real-world practices, right-turning vehicles are not regulated by traffic signals.

**Pressure** Pressure is a metric used to quantify the imbalance of traffic flow at an intersection, specifically between incoming and outgoing lanes. It is typically defined as the difference between the queue lengths (or vehicle densities) of incoming and outgoing lanes for a specific movement. The total pressure  $P$  for a phase is usually calculated as the sum of pressures over all movements controlled by that phase:  $P = \sum_{(i,j)} (q_i - q_j)$ , where  $q_i$  is the number of vehicles on the incoming lane  $l_{in}^i$ , and  $q_j$  is that on the corresponding outgoing lane  $l_{out}^j$ , as shown in Figure 2.

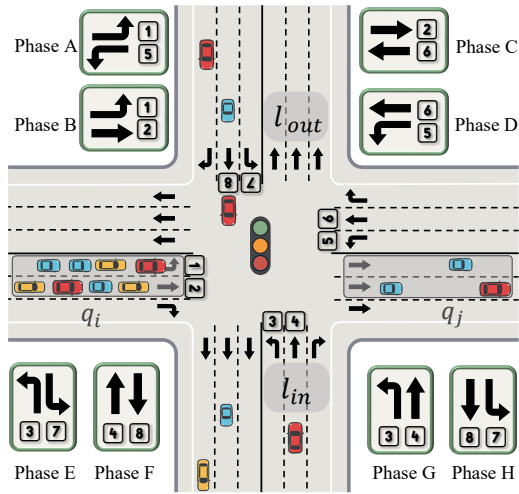


Figure 2: The illustration of intersection, phases and pressure

### 3.2 Problem Formulation

We model the large-scale TSC problem based on Multi-agent Markov Decision Process (MMDPs) as hierarchical MMDPs, which can be defined by the tuple  $\langle \mathcal{I}, \mathcal{S}, \mathcal{O}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ . In this paper, we refer to the high-level policy of HRL as the Meta-Policy, which is responsible for generating abstract goals or selecting sub-policies. The low-level policy, referred to as the Sub-Policy, executes concrete actions to fulfill the objectives specified by the Meta-Policy. The set of intersections is defined as  $\mathcal{I} = \{1, 2, \dots, N\}$ , where  $N$  is the total number of intersections and each  $i \in \mathcal{I}$  is controlled by an agent. All agents share the same set of parameters.  $S$  denotes the global state, and  $O$  is a partial observation derived from  $S$  via an observation function  $f_{obs} : s \rightarrow o$ , where  $s \in S, o \in O$ . The action space is defined as  $A = \{A^H, A^L\}$ , where  $A^H$  contains only a single global sub-goal  $a^H$  assigned by the Meta-Policy, and each Sub-Policy agent selects its local action  $A^L = \{a_i^L\}_{i \in \mathcal{I}}$  based on its local observation and the global sub-goal  $a^H$ .  $P$  and  $\gamma$  denote the transition probability function and the discount factor separately. The reward function  $\mathcal{R}$  is shared among all intersections and consists of a global high-level reward and local low-level rewards, formulated as:

$$\mathcal{R}(s, a^H, a_1^L, \dots, a_N^L, s') = \mathcal{R}^H(s, a^H, s') + \sum_{i \in \mathcal{I}} \mathcal{R}_i^L(s_i^L, a_i^L, s_i^L)$$

where  $s' \in \mathcal{S}$  is the next state generated according to the transition probability  $\mathcal{P}(s'|s, a^H, a^L)$ ,  $\mathcal{R}^H$  reflects the overall performance of the sub-goal assignment, and  $\mathcal{R}_i^L$  corresponds to the local objective of each agent  $i \in \mathcal{I}$ , such as minimizing queue length or waiting time. Correspondingly, the overall policy is factorized into a high-level global policy  $\pi^H$  and decentralized local policies  $\pi^L$ , such that the joint policy  $\pi^{all}$  is defined as the product of these two levels. Specifically, the high-level policy  $\pi^H$  selects a global sub-goal  $a^H$  based on the global state  $s$ , while low-level policy  $\pi^L$  chooses the local action  $a_i^L$  conditioned on the local observation and the global sub-goal  $a^H$ . The overall joint policy can thus be written as:

$$\pi^{all}(a^H, a^L | s) = \pi^H(a^H | s) \cdot \prod_{i=1}^N \pi^L(a_i^L | o^i, a^H)$$

All agents coordinate to maximize the expected cumulative discounted reward:

$$\max_{\pi^{all}} \mathbb{E}_{\pi^{all}} \left[ \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t^H, a_t^L) \right]$$

Note that, for clarity in illustrating the overall process of the Markov decision process, the observation function  $f_{obs}$  defined above has been simplified. In practical implementations, the observation functions for the Meta-Policy and the Sub-Policy workers are not identical. The detailed design and differences between these components are elaborated in the next section.

## 4 Methodology

In this section, we propose a novel HRL framework HiLight for traffic signal control. As shown in Figure 3, the architecture consists of two branches: a global Meta-Policy and a local Sub-Policy, which work in coordination while being jointly optimized in an adversarial manner to achieve efficient control across large-scale traffic networks.

To address the challenge of balancing global coordination and local adaptability in large-scale traffic environments, HiLight leverages a Meta-Policy, based on Transformer and LSTM, to extract global representations from historical regional observations, and a shared-parameter Sub-Policy to make localized control decisions based on both local states and global guidance. The Meta-Policy provides a goal that serves as a high-level directional signal for all agents, while the Sub-Policy adapts its behavior accordingly through an Actor-Critic framework. Meanwhile, to improve learning efficiency and policy performance under sparse rewards and multi-agent objectives, we introduce an adversarial training mechanism, in which the Meta-Policy aims to generate increasingly challenging goals, and the Sub-Policy seeks to surpass them. The following sections provide more details of the HiLight.

### 4.1 Meta-Policy

In large-scale TSC scenarios, due to the high dimensionality of the global state, traffic signals are typically controlled using MARL, where each intersection is managed by an individual agent

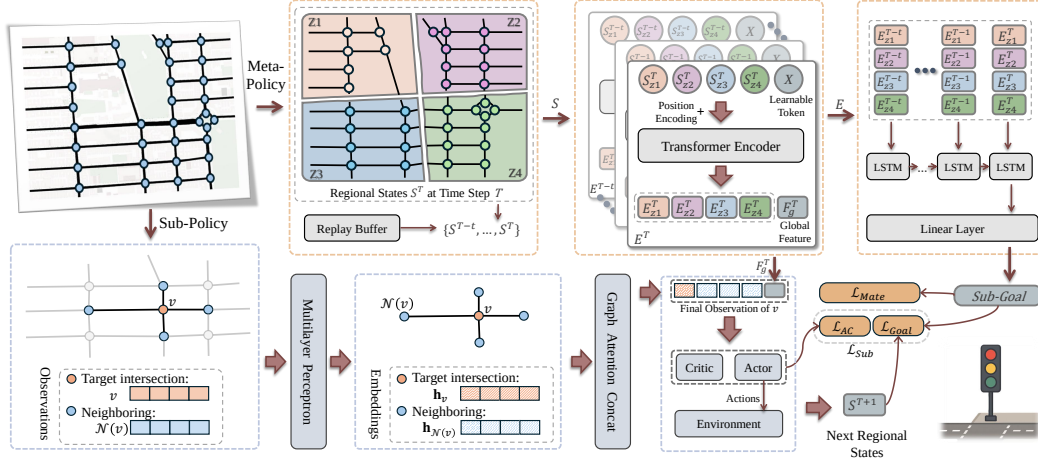


Figure 3: An overview of HiLight.

making local decisions. To enhance the perception range of each agent, existing studies mainly adopt approaches such as assigning collaborators or forming groups[38, 29]. However, in large-scale networks, the information provided by such methods remains limited. Agents often lack awareness of the overall traffic conditions across the network, which may lead to decisions that are misaligned with the global traffic efficiency. In contrast to these approaches, the Meta-Policy in HiLight operates at the global level, providing each agent with global information and guiding their optimization direction through sub-task assignments, thereby ensuring consistency with the overall objective of improving network-wide traffic flow.

#### 4.1.1 Transformer Encoding Across Subregions

After obtaining the sequential inputs for each subregion, Meta-Policy employs a Transformer encoder to enable information exchange across different subregions and extract global traffic patterns. Given the input data structured as  $(M+1, T, d_{reg})$ , where  $M$  is the number of subregions, and the additional dimension represents the prepended a learnable input token  $\mathbf{x}_{token} \in \mathbb{R}^d$ ,  $d_{reg}$  is the dimensionality of the regional feature representation. Specifically, the input to Transformer encoder for time step  $t$  is:

$$\mathbf{X}^t = [\mathbf{x}_{token}, \mathbf{s}_1^t, \mathbf{s}_2^t, \dots, \mathbf{s}_M^t] \in \mathbb{R}^{(M+1) \times d_{reg}}$$

where  $\mathbf{s}_z^t \in \mathbb{R}^d$  is the regional state at time step  $t$ , see Appendix A for more details of regional state. To incorporate positional information across the subregions, we apply sinusoidal positional encoding as  $\mathbf{X}_{pos}^t = \mathbf{X}^t + \mathbf{PE}(pos)$ , with the positional encoding defined as:

$$\mathbf{PE}(pos, 2i) = \sin(\frac{pos}{\tau^{2i/d_{model}}}), \quad \mathbf{PE}(pos, 2i+1) = \cos(\frac{pos}{\tau^{2i/d_{model}}})$$

where  $\tau$  is a scaling parameter. The transformer encoder processes the input and computes attention among subregions:

$$[\mathbf{E}_G^t, \mathbf{E}_1^t, \mathbf{E}_2^t, \dots, \mathbf{E}_M^t] = \text{TransformerEncoder}(\mathbf{X}_{pos}^t)$$

Here,  $\mathbf{E}_G^t$  denotes the embedding of the input token summarizing the global state at time step  $t$ , and  $\mathbf{E}_z^t$  represents the embedding of subregion  $z$  at time step  $t$ , each subregion embedding is calculated via self-attention across subregions. The global embedding  $\mathbf{E}_g = \mathbf{E}_G^T$  at the current time step  $T$  is directly provided to the sub-policy as one of the observations of agents, to enhance global context awareness during Sub-Policy decision-making.

#### 4.1.2 LSTM-based Sub-goal Generation

To produce sub-goals reflecting temporal dependencies and guide local agents with coherent long-term objectives, the subregion embeddings from the Transformer across the past  $T$  timesteps are reorganized and input into the LSTM network.

First, we transpose the embeddings to shape the data appropriately for temporal modeling:

$$\hat{\mathbf{E}}_z = [\mathbf{E}_z^1, \mathbf{E}_z^2, \dots, \mathbf{E}_z^T] \in \mathbb{R}^{T \times d_{reg}}, \quad z \in \{1, 2, \dots, M\}$$

Then, the embeddings are rearranged such that the subregions become the batch dimension and the temporal dimension is used as the recurrent axis. This reshaping yields an input of shape  $(T, M, d)$  for the LSTM:

$$\mathbf{H} = LSTM(\mathbf{E}_{1:T}, \dots, \mathbf{E}_{M:T}) \in \mathbb{R}^{T \times M \times d_{hidden}}$$

where the LSTM learns to model temporal dependencies within each subregion's trajectory. The final hidden state of each subregion is then aggregated to produce a global sub-goal representation:

$$\mathbf{G} = \phi([\mathbf{h}_1^T \parallel \mathbf{h}_2^T \parallel \dots \parallel \mathbf{h}_M^T]) \in \mathbb{R}^{d_g}$$

where  $\mathbf{h}_{z_i}^T$  denotes the final hidden state of the LSTM for subregion  $z_i$ ,  $\parallel$  represents concatenation, and  $\phi(\cdot)$  is a feedforward layer to project the concatenated features into a fixed-size global sub-goal vector  $\mathbf{G}$ . This vector  $\mathbf{G}$  is broadcasted to all agents of Sub-Policy and serves as a shared directional objective, encouraging the agents at each intersection to align their local decisions with global traffic optimization trends.

## 4.2 Sub-Policy

Meta-Policy provides both global representations and high-level optimization objectives, while the core responsibility of the Sub-Policy is to perform real-time traffic signal control decisions at each intersection based on comprehensive observation information. Each intersection is modeled as an independent agent, and all agents share the same network parameters. The Sub-Policy integrates local traffic observations from the surrounding neighbors with the global guidance representation provided by the Meta-Policy, thereby constructing a final observation with multi-scale information awareness. This final observation serves as the input to both the policy and value networks within an Actor-Critic framework.

### 4.2.1 Local Observation Encoding

At each time step, the local observation for a given intersection is composed of its own state and the states of its four nearest neighboring intersections, resulting in a total of five observation vectors. The information from neighboring intersections is dynamically weighted according to its relative importance at the current time step. This design captures both the temporal variability and directional sensitivity of traffic flow. For instance, during morning or evening rush hours, different neighbors may serve as upstream intersections that carry major incoming traffic.

Let  $\mathbf{o}_v^t \in \mathbb{R}^k$  denote the raw  $k$ -dimensional observation vector of intersection  $v$  at time  $t$ ; the specific observation features are detailed in the Appendix A. Let  $\mathcal{N}(v)$  denote the set of its four nearest neighboring intersections. At time  $t$ , each observation  $\mathbf{o}_v^t$  is first encoded into an intermediate representation  $\mathbf{h}_v^t$  via a shared Multilayer Perceptron (MLP).

### 4.2.2 Dynamic Neighbor Aggregation

These embedded vectors are directly passed to the Graph Attention Concat (GAC) module. Let  $\mathbf{H}^t = [\mathbf{h}_1^t, \dots, \mathbf{h}_N^t] \in \mathbb{R}^{N \times F}$  denote the feature matrix at time  $t$  for all  $N$  intersections, and let  $\mathbf{A} \in \mathbb{R}^{N \times N}$  be the adjacency matrix where  $A_{ij} = 1$  if node  $j$  is a neighbor of node  $i$ . For each valid edge  $(i, j)$  with  $A_{ij} = 1$ , we compute an attention score as:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{h}_i^t \parallel \mathbf{h}_j^t])$$

with  $\mathbf{a} \in \mathbb{R}^{2F}$  a learnable attention vector and  $\parallel$  denoting vector concatenation, and attention scores are computed as  $\alpha_{ij} = \text{softmax}(e_{ij})$  over the neighborhood  $\mathcal{N}_i = \{j \mid A_{ij} = 1\}$ . The final representation  $\mathbf{z}_i^t$  is constructed by concatenating  $\mathbf{h}_i^t$  with the weighted neighbor features:

$$\mathbf{z}_i^t = \mathbf{h}_i^t \parallel \alpha_{ij_1} \mathbf{h}_{j_1}^t \parallel \alpha_{ij_2} \mathbf{h}_{j_2}^t \parallel \alpha_{ij_3} \mathbf{h}_{j_3}^t \parallel \alpha_{ij_4} \mathbf{h}_{j_4}^t$$

If fewer than four neighbors exist, zero vectors are padded to maintain fixed output size  $\mathbf{z}_i^t \in \mathbb{R}^{5F}$ . Eventually, the representation  $\mathbf{z}_i^t$  is concatenated with the global feature vector  $F_g$  generated by the Meta-Policy to form the final observation of intersection  $i$ :  $\mathbf{o}_i^{final} = \mathbf{z}_i^t \parallel F_g$

## 4.3 Joint Optimization

In this section, we describe the joint optimization mechanism integrating Meta-Policy and Sub-Policy under an adversarial hierarchical reinforcement learning framework. We elaborate on reward definitions, trajectory formulation, and adversarial loss functions.

### 4.3.1 Reward and Trajectory

The reward structure optimizes global traffic efficiency and local intersection throughput simultaneously. At each time step  $t$ , each intersection (agent)  $i$  receives a local reward  $r_i^t$  that evaluates its own traffic condition. The reward is calculated as a weighted combination of several key metrics: queue length  $ql_i^t$ , average vehicle waiting time  $wt_i^t$ , delay time  $dt_i^t$ , pressure  $ps_i^t$  and speed score  $ss_i^t$ , with detailed definitions in Appendix A. The local reward is then given as:  $r_i^t = -(ql_i^t + wt_i^t + dt_i^t + ps_i^t - ss_i^t)$ .

Meanwhile, the Meta-Policy sets global goal vectors at each timestep  $G^t$ , comprising targets for global waiting time  $G_w^t$  and global queue length  $G_q^t$ . The global reward  $r_g^t$  assesses the discrepancy between the current global state, specifically total waiting time  $W_{global}^t$  and total queue length  $Q_{global}^t$ . Then, the goal reward is defined as:

$$r_g^t = -[\beta_q (W_{global}^t - G_q^t) + \beta_w (Q_{global}^t - G_w^t)]$$

where  $\beta$  is scalar weights balancing the importance of each metric. Based on these reward definitions, trajectories are formed at each timestep. For each intersection  $i$ , the trajectory is represented as  $\tau_i^t = (\mathbf{o}_i^{final,t}, a_i^t, r^t, \mathbf{o}_i^{final,t+1})$ , where  $r^t = r_i^t + r_g^t$ .

### 4.3.2 Adversarial Training Mechanism

In our framework, the Meta-Policy generates a global goal  $G^t$  intended to predict the global traffic status at timestep  $t$ , but deliberately adjusted to be slightly more optimal than the actual outcome. The Meta-Policy’s objective consists of two components: the first encourages  $G^t$  to approximate the actual global traffic condition after  $t$  steps, and the second incentivizes the Meta-Policy to set even more ambitious sub-goals by rewarding more challenging targets. Formally, the Meta-Policy minimizes the following loss:

$$\mathcal{L}_{Meta} = \mathbb{E}_{\tau \sim \pi_{sub}} [\|G^t - (W_{global}^t, Q_{global}^t)\|^2 + \eta_1 r_g^t]$$

Meanwhile, the Sub-Policy aims to not only maximize cumulative returns via standard Actor-Critic updates but also align its achieved global traffic performance with the targets set by the Meta-Policy. The Sub-Policy minimizes the following loss:

$$\mathcal{L}_{Sub} = \mathbb{E}_{\tau \sim \pi_{sub}} [\mathcal{L}_{AC} + \eta_2 (\beta_q (W_{global}^t - G_q^t) + \beta_w (Q_{global}^t - G_w^t))]$$

where  $\mathcal{L}_{AC}$  denotes the standard Actor-Critic loss (policy gradient + value loss) and  $\eta, \beta$  balance the importance of adversarial loss.

## 5 Experiments and Results

### 5.1 Experimental Setup

We utilize the Simulation of Urban MObility (SUMO) traffic simulator as the environment for all experiments, which is employed in a non-GUI mode to enable efficient large-scale simulations. Each episode is set to 3600 seconds of simulated time, with a rollout length of 240 steps. The yellow light duration is 5 seconds, and each signal phase iteration lasts 10 seconds. As shown in Table 1, our experimental setup includes one large-scale scenario based on Manhattan, containing 2668 traffic lights, as well as five commonly used benchmark scenarios.

For the Manhattan2668 scenario, we define the network using OSMWebWizard<sup>1</sup> and manually refine its intersection details. Traffic flow is generated based on open-source taxi trip data<sup>2</sup>, and we select dates associated with adverse weather conditions using Local Climatological Data<sup>3</sup> to simulate the Adverse Weather Flow. In addition, we simulate two other realistic traffic patterns based on actual urban traffic conditions: the Peak Transition Flow, which represents a sudden influx of vehicles during the transition from off-peak to peak hours, and the Holiday Rush Flow, which reflects surges in traffic during holiday periods with increased travel demand. Visualizations of the traffic scenarios described in Appendix D. Ablation Study can be found in the Appendix B.

<sup>1</sup><https://sumo.dlr.de/docs/Tutorials/OSMWebWizard.html>

<sup>2</sup><https://opendata.cityofnewyork.us/data>

<sup>3</sup><https://www.ncei.noaa.gov/products/land-based-station/local-climatological-data>

Table 1: Statistics of evaluation scenarios

| Scenarios       | Cologne8     | Grid4×4     | Arterial4×4 | Ingolstadt21 | Grid5×5     | Manhattan2668   |                 |               |
|-----------------|--------------|-------------|-------------|--------------|-------------|-----------------|-----------------|---------------|
|                 |              |             |             |              |             | Config 1        | Config 2        | Config 3      |
| Country         | Germany      | Synthetic   | Synthetic   | Germany      | Synthetic   | United States   | United States   | United States |
| # Intersections | 8            | 16          | 16          | 21           | 25          | 2668            | 2668            | 2668          |
| Min flow (/s)   | 0.037        | 0.018       | 0.026       | 0.039        | 0.033       | 1.389           | 1.944           | 4.167         |
| Max flow (/s)   | 0.059        | 0.038       | 0.185       | 0.209        | 0.379       | 6.944           | 5.139           | 7.639         |
| Flow type       | Morning peak | Multi-modal | Multi-modal | Morning peak | Multi-modal | Peak Transition | Adverse Weather | Holiday Rush  |
|                 | Real flow    | Gaussian    | Gaussian    | Real flow    | Gaussian    | Real flow       | Real flow       | Real flow     |

Table 2: Performance comparison on Manhattan2668 under three traffic flow settings

| Methods               | Avg. Trip Time      |                      |                      | Avg. Delay Time      |                     |                     |
|-----------------------|---------------------|----------------------|----------------------|----------------------|---------------------|---------------------|
|                       | Peak Transition     | Adverse Weather      | Holiday Rush         | Peak Transition      | Adverse Weather     | Holiday Rush        |
| FTC (2004)            | 1474.83 ±11.45      | 1942.79 ±5.14        | 2116.07 ±10.81       | 1385.62 ±24.99       | 1530.38 ±29.6       | 1517.91 ±19.78      |
| MaxPressure (2014)    | 1175.25 ±8.35       | 1481.05 ±9.34        | 1667.4 ±7.09         | 975.95 ±14.13        | 1092.04 ±11.00      | 1133.57 ±18.29      |
| CoLight (2019)        | 907.43 ±0.00        | 1194.92 ±0.00        | 1313.62 ±0.00        | 713.15 ±0.00         | 793.59 ±0.00        | 783.52 ±0.00        |
| MPLight (2020)        | 1111.42 ±23.52      | 1463.75 ±10.37       | 1549.02 ±21.5        | 800.14 ±1.59         | 887.25 ±15.4        | 907.16 ±7.78        |
| MetaLight (2020)      | 823.6 ±24.37        | 1085.5 ±89.04        | 1178.03 ±86.22       | <u>567.55 ±33.01</u> | 752.64 ±34.57       | 746.17 ±40.72       |
| IPPO (2021)           | 981.12 ±58.13       | 1486.93 ±75.09       | 1411.91 ±82.16       | 794.88 ±16.84        | 926.85 ±14.58       | 901.33 ±10.01       |
| rMAPPO (2022)         | 1140.17 ±35.26      | 1338.17 ±12.07       | 1568.61 ±26.50       | 863.48 ±33.89        | 904.72 ±25.77       | 901.33 ±11.40       |
| MetaGAT (2022)        | 772.92 ±0.00        | 1024.78 ±0.00        | 1127.09 ±0.00        | 681.61 ±0.00         | 761.8 ±0.00         | 756.24 ±0.00        |
| GESA (2024)           | <u>732.28 ±7.52</u> | <u>930.47 ±18.72</u> | 1181.53 ±17.65       | 673.82 ±5.39         | 723.29 ±8.39        | <u>629.82 ±7.74</u> |
| CoSLight (2024)       | 783.15 ±14.96       | 1094.27 ±34.30       | 1284.27 ±28.87       | 586.93 ±12.04        | 728.94 ±9.37        | 662.32 ±10.94       |
| X-Light (2024)        | 843.36 ±0.00        | 1103.85 ±0.00        | <u>1051.61 ±0.00</u> | 634.07 ±0.00         | <u>667.22 ±0.00</u> | 641.89 ±0.00        |
| <b>HiLight (ours)</b> | <b>690.88 ±9.01</b> | <b>913.84 ±14.9</b>  | <b>980.24 ±20.44</b> | <b>549.02 ±6.24</b>  | <b>649.71 ±1.40</b> | <b>598.65 ±9.16</b> |

## 5.2 Evaluation Metrics and Compared Methods

We employ two widely used evaluation metrics in TSC: Average Travel Time (ATT) and Average Delay Time (ADT). ATT measures the average duration that all vehicles spend in the scenario, from entering to exiting. ADT captures the average delay caused by congestion or traffic signals.

To comprehensively evaluate the performance of HiLight, we compare it with eleven existing methods, including two traditional FTC[1] and MaxPressure[2]—as well as eight state-of-the-art RL or MARL algorithms: CoLight[27], MPLight[39], MetaLight[30], IPPO[40], rMAPPO[26], MetaGAT[41], GESA[14], CoSLight[42] and X-Light[43]. Detailed descriptions of these methods are provided in the AppendixC.

## 5.3 Performance in Large-Scale Scenario

In a large-scale traffic network based on the Manhattan map, our approach achieves the lowest mean trip times (690.88 s, 913.84 s, 980.24 s) and delay times (549.02 s, 649.71 s, 598.65 s) across all traffic flow settings, as shown in Table 2, outperforming even the strongest baselines by **2.6–6.8%**.

These results demonstrate the effectiveness of our hierarchical structure in handling large and dynamic traffic systems. Compared to flat or fully decentralized methods such as CoLight or IPPO, our method benefits from its two-level decision-making mechanism. The high-level policy captures global traffic dynamics and provides strategic guidance to each region, which helps mitigate the lack of coordination observed in multi-agent methods. Meanwhile, the low-level policies, conditioned on regional objectives, allow for fine-grained control tailored to local traffic states. The combination of top-down coordination and bottom-up adaptability allows our method to generalize better under highly dynamic conditions such as weather disruptions or holiday rushes, where other methods often struggle with delayed responses or local optima.

## 5.4 Performance in Standard Benchmark Scenarios

To further evaluate the generalizability of our method, we conduct experiments on a variety of traffic scenarios with different network sizes, ranging from small-scale networks such as Cologne8 (8 traffic lights) and Grid4×4 (16 traffic lights), to more complex ones such as Manhattan2668, which includes

Table 3: Performance comparison on standard scenarios and Manhattan2668

| Methods                  | Cologne8                          | Grid4x4                            | Arterial4x4                        | Ingolstadt21                       | Grid5x5                            | Manhattan2668                       |
|--------------------------|-----------------------------------|------------------------------------|------------------------------------|------------------------------------|------------------------------------|-------------------------------------|
| Avg. Trip Time(seconds)  |                                   |                                    |                                    |                                    |                                    |                                     |
| FTC                      | 124.40 $\pm$ 1.99                 | 206.68 $\pm$ 0.54                  | 828.38 $\pm$ 8.17                  | 319.41 $\pm$ 24.48                 | 550.38 $\pm$ 8.31                  | 1844.56 $\pm$ 9.13                  |
| MaxPressure              | 95.96 $\pm$ 1.11                  | 175.97 $\pm$ 0.70                  | 686.12 $\pm$ 9.57                  | 375.25 $\pm$ 2.40                  | 274.15 $\pm$ 15.23                 | 1441.23 $\pm$ 8.26                  |
| CoLight                  | 89.72 $\pm$ 0.00                  | 163.52 $\pm$ 0.00                  | 409.93 $\pm$ 0.00                  | 337.46 $\pm$ 0.00                  | 242.37 $\pm$ 0.00                  | 1138.66 $\pm$ 0.00                  |
| MPLight                  | 98.44 $\pm$ 0.62                  | 179.51 $\pm$ 0.95                  | 541.29 $\pm$ 45.24                 | 319.28 $\pm$ 10.48                 | 261.76 $\pm$ 6.60                  | 1374.73 $\pm$ 18.46                 |
| MetaLight                | 91.57 $\pm$ 0.75                  | 169.21 $\pm$ 1.26                  | 381.77 $\pm$ 12.85                 | 292.26 $\pm$ 4.40                  | 247.83 $\pm$ 5.99                  | 1029.04 $\pm$ 66.54                 |
| IPPO                     | 90.87 $\pm$ 0.40                  | 167.62 $\pm$ 2.42                  | 431.31 $\pm$ 28.55                 | 379.22 $\pm$ 34.03                 | 259.28 $\pm$ 9.55                  | 1293.32 $\pm$ 71.79                 |
| rMAPPO                   | 97.68 $\pm$ 2.03                  | 164.96 $\pm$ 1.87                  | 565.67 $\pm$ 44.8                  | 453.61 $\pm$ 29.66                 | 300.9 $\pm$ 8.31                   | 1348.98 $\pm$ 24.61                 |
| MetaGAT                  | 96.74 $\pm$ 0.00                  | 165.23 $\pm$ 0.00                  | 374.8 $\pm$ 0.87                   | 290.73 $\pm$ 0.45                  | 266.6 $\pm$ 0.00                   | 974.93 $\pm$ 0.00                   |
| GESA                     | 90.59 $\pm$ 0.74                  | 161.33 $\pm$ 1.34                  | 393.57 $\pm$ 13.72                 | 320.02 $\pm$ 5.57                  | 252.11 $\pm$ 9.94                  | 948.09 $\pm$ 14.63                  |
| CoSLight                 | 90.46 $\pm$ 0.55                  | 159.11 $\pm$ 3.12                  | 364.21 $\pm$ 4.78                  | 284.62 $\pm$ 2.57                  | 220.32 $\pm$ 1.71                  | 1053.90 $\pm$ 26.04                 |
| X-Light                  | <b>88.55 <math>\pm</math>0.00</b> | 162.47 $\pm$ 0.00                  | 349.6 $\pm$ 0.00                   | 278.05 $\pm$ 0.00                  | 220.63 $\pm$ 0.00                  | 999.61 $\pm$ 0.00                   |
| HiLight                  | 90.83 $\pm$ 0.34                  | <b>159.07 <math>\pm</math>0.97</b> | <b>341.42 <math>\pm</math>4.86</b> | <b>272.50 <math>\pm</math>6.03</b> | <b>204.26 <math>\pm</math>8.14</b> | <b>861.65 <math>\pm</math>14.78</b> |
| Avg. Delay Time(seconds) |                                   |                                    |                                    |                                    |                                    |                                     |
| FTC                      | 62.38 $\pm$ 2.95                  | 94.64 $\pm$ 0.43                   | 1234.30 $\pm$ 6.50                 | 183.70 $\pm$ 26.21                 | 790.18 $\pm$ 7.96                  | 1477.97 $\pm$ 24.79                 |
| MaxPressure              | 31.93 $\pm$ 1.07                  | 64.01 $\pm$ 0.71                   | 952.53 $\pm$ 12.48                 | 275.36 $\pm$ 14.38                 | 240.00 $\pm$ 18.43                 | 1067.19 $\pm$ 14.47                 |
| CoLight                  | 25.56 $\pm$ 0.00                  | 51.58 $\pm$ 0.00                   | 776.61 $\pm$ 0.00                  | 226.06 $\pm$ 0.00                  | 248.32 $\pm$ 0.00                  | 763.42 $\pm$ 0.00                   |
| MPLight                  | 34.38 $\pm$ 0.63                  | 67.52 $\pm$ 0.97                   | 1083.18 $\pm$ 63.38                | 185.04 $\pm$ 10.70                 | 213.78 $\pm$ 14.44                 | 864.85 $\pm$ 8.26                   |
| MetaLight                | 27.61 $\pm$ 0.78                  | 57.56 $\pm$ 0.76                   | 862.32 $\pm$ 39.01                 | 164.80 $\pm$ 3.75                  | 209.13 $\pm$ 19.40                 | 688.79 $\pm$ 36.1                   |
| IPPO                     | 26.82 $\pm$ 0.43                  | 56.38 $\pm$ 1.46                   | 914.58 $\pm$ 36.90                 | 247.68 $\pm$ 35.33                 | 243.58 $\pm$ 9.29                  | 874.35 $\pm$ 13.81                  |
| rMAPPO                   | 33.37 $\pm$ 1.97                  | 53.65 $\pm$ 1.00                   | 1185.20 $\pm$ 167.48               | 372.20 $\pm$ 39.85                 | 346.78 $\pm$ 28.25                 | 889.84 $\pm$ 23.69                  |
| MetaGAT                  | 26.85 $\pm$ 0.00                  | 50.32 $\pm$ 0.00                   | 772.36 $\pm$ 0.00                  | 176.86 $\pm$ 2.37                  | 234.8 $\pm$ 0.00                   | 733.22 $\pm$ 0.00                   |
| GESA                     | 26.50 $\pm$ 0.87                  | 49.60 $\pm$ 0.71                   | 775.22 $\pm$ 8.63                  | 209.57 $\pm$ 3.32                  | 210.74 $\pm$ 13.56                 | 675.64 $\pm$ 7.17                   |
| CoSLight                 | 24.79 $\pm$ 1.23                  | <b>46.11 <math>\pm</math>1.21</b>  | 744.98 $\pm$ 16.49                 | 179.73 $\pm$ 14.26                 | 178.54 $\pm$ 4.34                  | 659.04 $\pm$ 10.78                  |
| X-Light                  | <b>24.31 <math>\pm</math>0.00</b> | 50.27 $\pm$ 0.00                   | 697.79 $\pm$ 0.00                  | 160.39 $\pm$ 0.00                  | 187.74 $\pm$ 0.00                  | 647.73 $\pm$ 0.00                   |
| HiLight                  | 24.81 $\pm$ 0.93                  | <u>48.25 <math>\pm</math>0.53</u>  | <b>682.86 <math>\pm</math>8.95</b> | <b>155.70 <math>\pm</math>1.56</b> | <b>164.92 <math>\pm</math>3.70</b> | <b>599.13 <math>\pm</math>5.60</b>  |

up to 2668 intersections. The results, summarized in Table 3, reveal a clear trend: while our method performs competitively in small-scale networks, its advantages become increasingly pronounced as the network size grows. As visualized in Figure 4, this advantage scales positively with the number of intersections in the scenario. Note that the results of Manhattan2668 in Table 3 is calculated as the average over the three traffic flow settings as reported in Table 2.

## 6 Conclusion

In this work, we proposed HiLight, a hierarchical reinforcement learning framework that incorporates global adversarial guidance for large-scale traffic signal control. Our method combines a Transformer-LSTM-based Meta-Policy with decentralized Sub-Policy to effectively balance global coordination and local adaptability. Extensive experiments across synthetic and real-world traffic scenarios demonstrate that HiLight consistently outperforms both traditional and state-of-the-art baselines, especially under dynamic and large-scale urban conditions.

**Limitation** Despite its promising performance, HiLight has several limitations. First, the current framework assumes full observability of global traffic states. However, in real-world scenarios, infrastructure failures may lead to missing or abnormal data at certain intersections, which can negatively impact model effectiveness. Second, although this study is the first to evaluate a hierarchical RL framework under diverse and realistic traffic scenarios, it still relies on simulated environments. Such simulations cannot fully capture the uncertainty and irregularities of real-world traffic systems, including incidents, construction activities, or emergency events. Future work may focus on improving scalability under partially observable conditions, incorporating real-time adaptive sensing mechanisms, and advancing deployment in real urban environments.

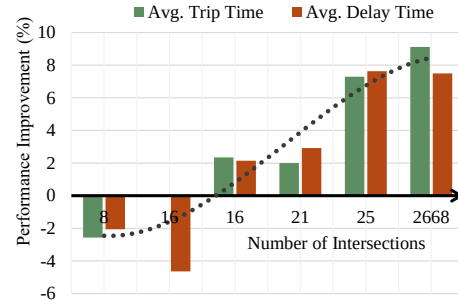


Figure 4: Improvement of HiLight over baseline across increasing intersection counts

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## A Detailed Descriptions and Computation of Observations

In our traffic signal control framework, the agent’s decision-making relies on a set of observations that capture the real-time traffic state at each intersection. These observations reflect critical aspects of traffic flow, congestion, and signal efficiency, enabling the model to respond dynamically to varying traffic conditions. Table 4 presents a detailed summary of each observation, including its semantic meaning and the method used for computation. In addition, the regional state includes three components: the "stop\_car\_num" and "waiting\_time" within the region, and the spatial coordinates of the subregion.

Table 4: Detailed descriptions and computation of observations

| Observation   | Meaning                                                                                                        | Computation                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| car_num       | Total number of vehicles on a lane, indicating the overall traffic density at the intersection.                | The sum of all vehicles on each lane.                                                                    |
| queue_length  | Length of the queue on incoming lanes, reflecting congestion or signal-induced queuing.                        | Queue length in meters; represented as an N-dimensional vector, where N is the number of incoming lanes. |
| occupancy     | Lane occupancy ratio, indicating the proportion of the lane occupied by vehicles.                              | The number of vehicles divided by lane capacity, per lane; N-dimensional.                                |
| flow          | Number of vehicles passing through per unit time, measuring traffic throughput.                                | The total number of vehicles passing through the intersection per unit time.                             |
| stop_car_num  | Number of stationary vehicles on a lane (e.g., stopped due to red lights or congestion).                       | The number of vehicles with speed $< 0.1m/s$ , normalized by lane capacity; N-dimensional.               |
| waiting_time  | Waiting time of the first vehicle on each incoming lane, reflecting delay at the intersection.                 | Sum of waiting times for the first vehicle on each incoming lane; N-dimensional.                         |
| average_speed | Average speed of vehicles passing through the intersection, reflecting potential delays due to signal changes. | The average speed of all vehicles passing through.                                                       |
| pressure      | Traffic pressure in a specific direction, reflecting imbalance between demand and discharge capacity.          | The difference in vehicle count between incoming and corresponding outgoing lanes.                       |
| delay_time    | The time loss due to the vehicle not reaching its expected speed.                                              | The difference between the ideal travel time and the actual travel time.                                 |

## B Ablation Study

To investigate the contribution of each core component in the proposed HiLight framework, an ablation study is conducted by progressively removing key modules and evaluating the resulting performance during training. As shown in the Episode–Reward curves in Figure 5a, the complete HiLight model achieves the best overall performance, exhibiting both the fastest convergence speed and the most stable behavior after convergence.

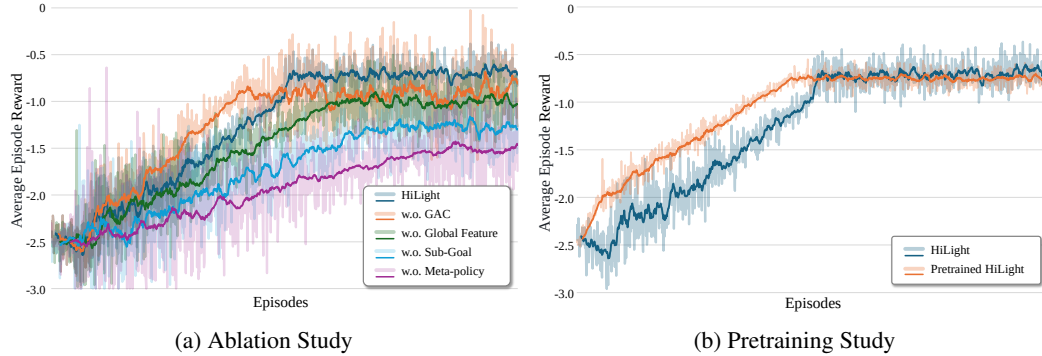


Figure 5: The episode–reward curves for ablation and pretraining analysis of the HiLight

The variant without the GAC module in the Sub-Policy (w.o. GAC) shows a slight decrease in performance. While convergence remains fast and final reward ranks second, the absence of neighbor features results in increased oscillations after convergence, indicating reduced local coordination stability. Excluding either the Global Feature (w.o. Global Feature) or the Sub-Goal mechanism (w.o. Sub-Goal) causes further degradation. Without global context, the agent’s ability to capture long-range dependencies is weakened, resulting in slower and less stable convergence. The absence

of sub-goals disables adversarial training and removes hierarchical guidance, leading to slower learning and significantly lower final rewards. The most severe performance loss is observed when removing the entire Meta-Policy (w.o. Meta-Policy), which eliminates both global feature modeling and sub-goal generation.

We also examine the impact of pretraining the Meta-Policy prior to full model training, as shown in Figure 5b. The Episode–Reward curve shows a smoother and faster start, without the initial reward drop observed in the baseline. Pretraining allows the Meta-Policy to generate more informative sub-goals in early training, which effectively guides the Sub-Policy toward better optimization directions and improves exploration efficiency. Although the learning slope becomes gentler in later stages, convergence speed is slightly improved, and the final performance remains on par with the non-pretrained HiLight.

## C Baseline Methods

To support reproducibility and provide further clarity, we present brief descriptions of all baseline methods compared with HiLight. These include two classical control algorithms and eight reinforcement learning-based methods. The classical baselines—FTC and MaxPressure—represent widely adopted rule-based strategies in traffic signal control. The remaining eight methods are state-of-the-art RL approaches that have been proposed to improve adaptability and scalability in dynamic traffic environments. Each method is briefly summarized below with reference to its original implementation and core characteristics.

### Conventional Methods:

- **Fix Time Control** (2004)[1] is a rule-based method that cycles through signal phases using a predetermined schedule without adapting to real-time traffic conditions.
- **MaxPressure** (2014)[2] adjusts signal phases based on queue lengths at each intersection, selecting the phase with the highest pressure to maximize vehicle throughput.

### RL-based Methods:

- **CoLight** (2019)[27] uses graph attention networks to enable adaptive communication among intersections, allowing each traffic signal to incorporate spatial-temporal influences from neighbors for improved coordination in large-scale networks.
- **MPLight** (2020)[39] leverages pressure-based reinforcement learning and carefully designed rewards to enable scalable and implicitly coordinated traffic signal control across large-scale urban networks.
- **MetaLight** (2020)[30] applies gradient-based meta-reinforcement learning to enable fast adaptation across traffic scenarios by transferring knowledge from previously learned environments, improving both learning speed and performance stability.
- **IPPO** (2021)[40] assigns an independent PPO agent to each intersection, where all agents share the same architecture but maintain separate model parameters.
- **rMAPPO** (2022)[26] deploys independent PPO agents at each intersection and trains them collectively using shared data from all intersections, enabling coordinated optimization of overall traffic efficiency.
- **MetaGAT** (2022)[41] combines context-based meta-learning with Graph Attention Networks to improve generalization across traffic environments while enhancing coordination and training efficiency.
- **GESA** (2024)[14] introduces a scenario-agnostic framework that unifies intersection representation and co-trains across multiple cities, enabling strong generalization and zero-shot transfer without manual annotation.
- **CoSLight** (2024)[42] enhances multi-intersection coordination by learning a separate collaborator selection policy that dynamically identifies effective teammates beyond immediate neighbors based on real-time intersection features.

- **X-Light** (2024)[43] employs a dual-level Transformer architecture to aggregate local and cross-city decision trajectories, enabling strong generalization and transferability across diverse urban traffic scenarios.

## D Visualizations of the scenarios

Figure 6, Figure 7, and Figure 8 illustrate the traffic scenarios used in our experiments.

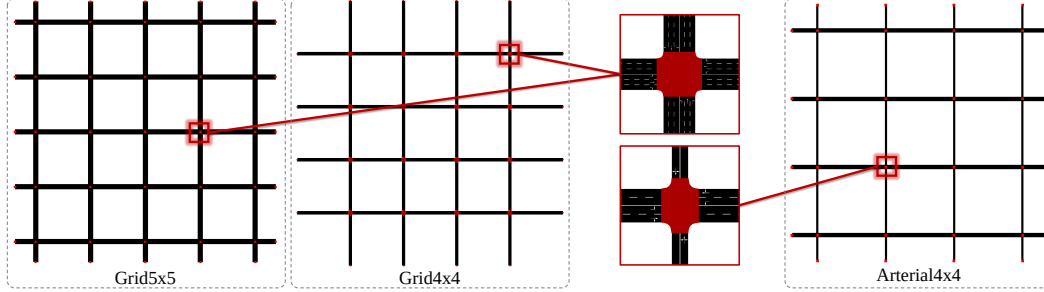


Figure 6: Synthetic scenarios with sample intersections.



Figure 7: German real-world scenarios with sample intersections

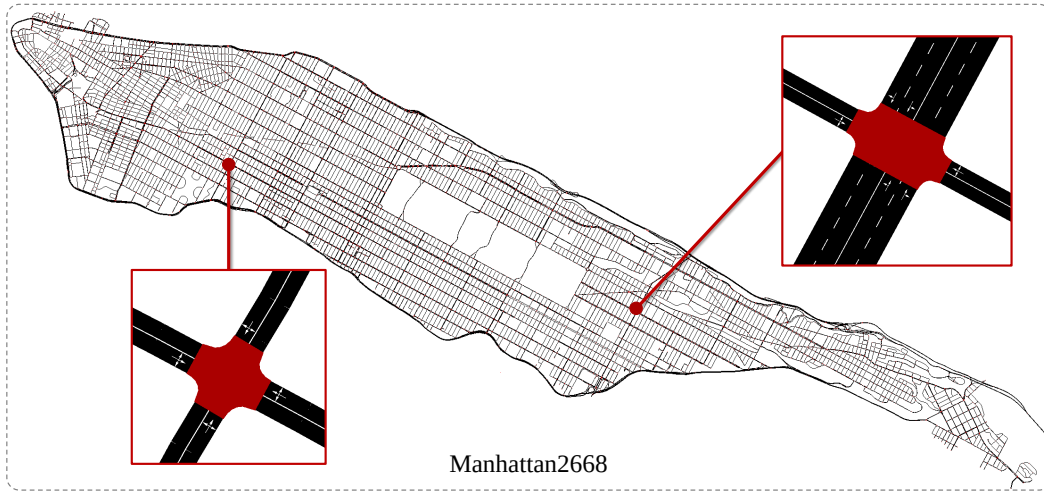


Figure 8: Manhattan real-world scenario with sample intersections

## E Hyper-parameter Settings

Table 5 summarizes the key hyperparameters used in our experiments. We train our model using Adam optimizer on NVIDIA GeForce RTX 4090 for around 19 hours and test it on the same GPU. When dividing the network into subregions using a grid-based approach, the number of intersections per region is generally kept within the order of  $10^1$  to ensure training efficiency. For example, in the Manhattan scenario, which contains a total of 2668 intersections and features a long and narrow layout, we partition the network into a  $4 \times 15$  grid, resulting in 60 regions with an average of approximately 45 intersections per region.

Table 5: Key hyperparameters used in experiments

| Description            | Value    | Description               | Value |
|------------------------|----------|---------------------------|-------|
| Total Steps            | 380000   | Hidden Layer Size         | 64    |
| Clipping Parameter     | 0.2      | Output Layer Gain         | 0.01  |
| Entropy Coefficient    | 0.01     | Number of Actions         | 8     |
| Value Loss Coefficient | 1.0      | Observation Dimension     | 66    |
| Maximum Gradient Norm  | 10.0     | Episode Duration (s)      | 3600  |
| Discount Factor        | 0.99     | Yellow Light Duration (s) | 5     |
| GAE Lambda             | 0.95     | Control Interval (s)      | 10    |
| Learning Rate          | $3e - 4$ | SUMO Start Port           | 15000 |

## F Network Architechure

The architecture, input format, processing procedure, and output of each module in HiLight are summarized in Table 6.

## G Evaluation Metrics

The calculation formulas of ATT and ADT are as follows:

$$ATT = \frac{1}{V} \sum_{i=1}^V (t_i^l - t_i^e), ADT = \frac{1}{V} \sum_{i=1}^V (T_i^r - T_i^t)$$

where  $t_i^e$  and  $t_i^l$  represent the entry and exit times of the  $i$ -th vehicle, respectively.  $T_i^r$  denotes the actual travel time of the  $i$ -th vehicle (i.e.,  $T_i^r = t_i^l - t_i^e$ ), and  $T_i^t$  represents the theoretical travel time of the  $i$ -th vehicle assuming it travels at the allowed speed without any delay.

## H Broad Impacts

HiLight has the potential to contribute positively to urban mobility and sustainability. By improving traffic efficiency, it can reduce fuel consumption, lower greenhouse gas emissions, and alleviate commuter stress. However, deploying such systems at scale also raises societal and ethical considerations. For example, data-driven traffic optimization may inadvertently favor high-density or economically privileged areas if not carefully monitored. Moreover, heavy reliance on historical data may reproduce existing biases in urban planning. Ensuring fairness, transparency, and inclusivity in model design and deployment will be essential. We encourage further interdisciplinary collaborations to explore these broader implications in depth.

Table 6: Detailed architecture and data flow of each component in HiLight

| Module Name                     | Purpose                                                                          | Architecture                                                                                | Input                                                         | Processing & Output                                                                                                                                                                                                      |
|---------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Meta-Policy Transformer Encoder | Extract global and regional spatiotemporal features across sub-regions           | Transformer Encoder: 3 layers, 2 heads, hidden size 4, feedforward dim = 165, no layer norm | (20, 4, 4): 20 steps, 4 regions, 4-dim observation per region | Adds a class token and positional encoding $\rightarrow$ Transformer encoder $\rightarrow$ output (5, 20, 4) $\rightarrow$ extract (1, 1, 4) as global feature; transform remaining (4, 20, 4) $\rightarrow$ (1, 20, 16) |
| Meta-Policy MultiLayer LSTM     | Capture global temporal patterns across all regions                              | 4-layer LSTM + FC layer                                                                     | (1, 20, 16): sequence of flattened regional features          | LSTM (hidden size = 256) $\rightarrow$ use last-step output $\rightarrow$ FC layer $\rightarrow$ (1, 1, 16) $\rightarrow$ reshape to (1, 4, 4)                                                                           |
| Graph Attention Concat Layer    | Aggregate local traffic context for each signal via attention over its neighbors | MLP + Graph Attention + Feature Concatenation                                               | (1, 16, 66): 16 nodes, each with 66-dim observation           | Linear projection $\rightarrow$ attention with 4 neighbors $\rightarrow$ weighted aggregation $\rightarrow$ concatenate self and neighbors $\rightarrow$ output (1, 16, 330)                                             |
| Feature Fusion                  | Combine global and local features as input for shared policy network             | Broadcast + Concatenation                                                                   | Global: (1, 1, 4), Local: (1, 16, 330)                        | Broadcast global feature to (1, 16, 4) $\rightarrow$ concatenate with local features $\rightarrow$ fused tensor (1, 16, 334)                                                                                             |
| Shared Head for Critic & Actor  | Project fused traffic features into shared latent representation                 | FC layers: 334 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 114 (with dropout)         | (1, 16, 334): concatenated per-node features                  | Three-layer MLP with ReLU and dropout $\rightarrow$ output (1, 16, 114) $\rightarrow$ used by actor and critic                                                                                                           |
| Critic Model                    | Estimate state-value and predict latent planning codes                           | Dual-branch MLP with partial feature fusion                                                 | (16, 114): shared feature per signal                          | Branch 1: 114 $\rightarrow$ 57 $\rightarrow$ concat $\rightarrow$ 1 $\Rightarrow$ value output (16, 1); Branch 2: 114 $\rightarrow$ 57 $\rightarrow$ 56 $\Rightarrow$ latent plan (16, 56)                               |
| Actor Model                     | Output traffic signal control actions                                            | 2-layer MLP with ReLU activation                                                            | (16, 114): shared feature per signal                          | 114 $\rightarrow$ 57 $\rightarrow$ 8 $\Rightarrow$ action logits (16, )                                                                                                                                                  |